

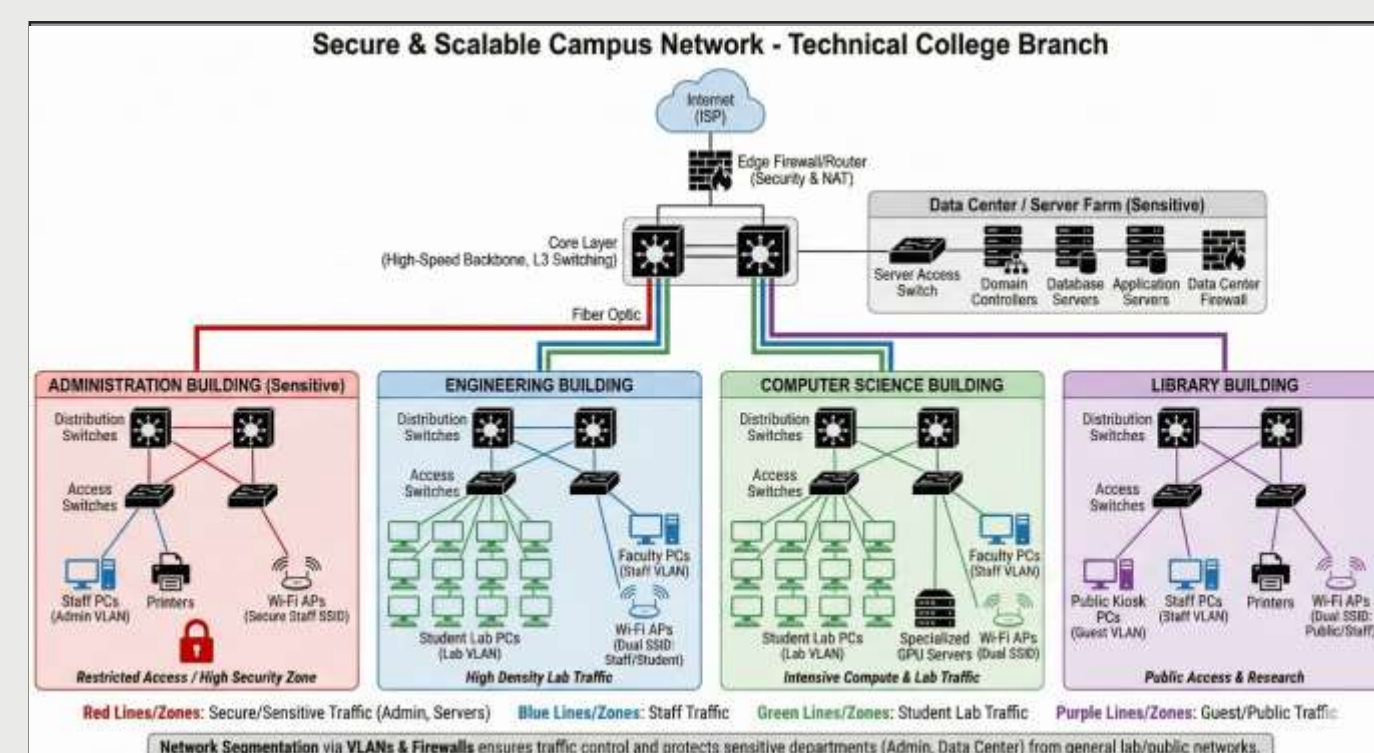
A New Technical College Campus Scalable & Secure Campus Network Design

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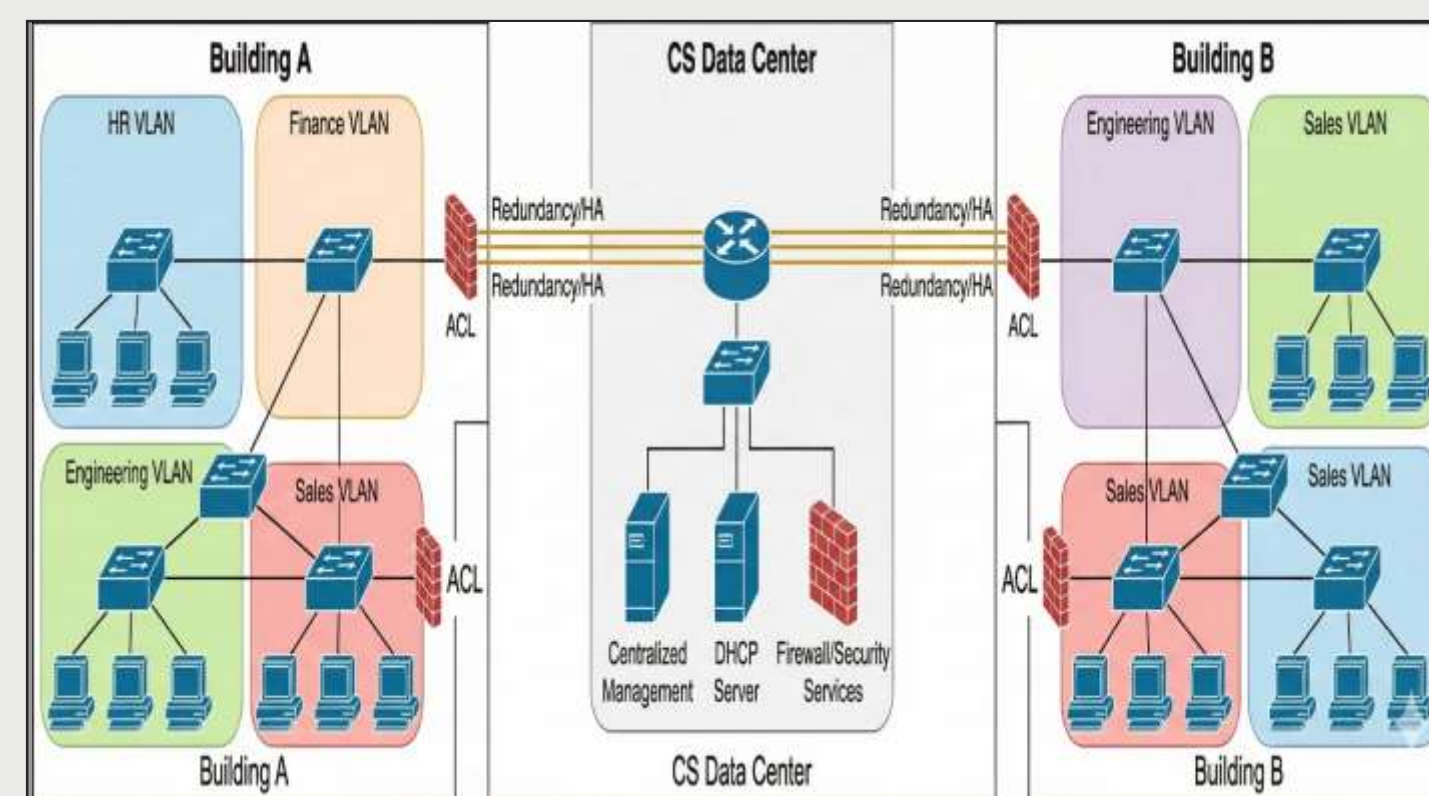
Problem

- Need a scalable and secure campus network for a new technical college branch.
- 4 buildings: Administration, Engineering, Computer Science, Library.
- Support staff offices, student labs, servers/services, and campus Wi-Fi.
- Must control traffic and protect sensitive departments.



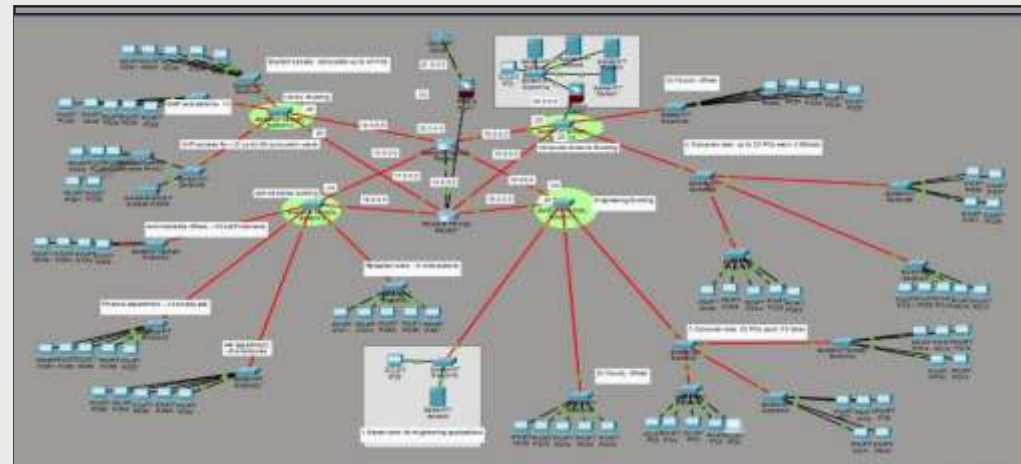
Goal

- High availability and redundancy between buildings.
- Segmentation per department using subnetting/VLANs.
- Centralized services and management in CS Data Center.
- Strong security controls (ACLs, SSH, firewalls).
- Easy IP management using DHCP.



Approach

- Router per building for segmentation and inter-building routing.
- Mesh topology backbone for redundancy and reliability.
- Class A private addressing for future expansion + structured subnetting.
- OSPF Area 0 across all routers for dynamic routing (justify: scalability, fast convergence, standard).
- Multilayer switches for speed, inter-VLAN routing, and improved security.
- DHCP for all networks (better than static IP administration).
- Include a small boxed subsection titled: **Alternative Low-Cost Design (Summary)**
 - Reduce high-end devices, keep VLANs + ACLs + OSPF
 - Simplify core architecture where possible
 - Maintain security around Data Center



Technologies used

- OSPF (Area 0)
- VLANs + subnetting
- ACLs (especially strict rules for Administration)
- SSH (secure management; avoid Telnet)
- Firewalls (protect Data Center, prevent external/cloud attacks)
- DHCP
- Wireless Access Points (campus Wi-Fi)
- Multilayer switches



Results and Conclusion

- Scalable addressing and subnetting allows growth.
- Mesh backbone improves reliability and reduces downtime.
- OSPF simplifies routing and adapts quickly to failures.
- ACLs, SSH, and firewalls strengthen security (Admin + Data Center protected).
- Design supports both wired labs and campus wireless users.
- Mention cost-aware alternative design availability. **IMPORTANT:** In this section, add a small "Cost Summary" callout:
 - Premium Design Total: **\$392,000**
 - Cost-Optimized Design Total: **\$235,000**
 - Savings: **\$157,000 (~40%)**
- One-line note about the key trade-off: less redundancy (star core), more centralized routing.

